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Solar roofing system

Abstract

A system includes a photovoltaic module having photovoltaic cells, each having a width, and a roofing shingle having an exposure zone and a headlap zone. A plurality of slots extends from the exposure zone to the headlap zone and define tooth portions. A first one of the tooth portions has a first side defined by a first slot and a second side defined by a second slot adjacent to the first slot. The first tooth portion has a first width that is the photovoltaic cell width multiplied by a first positive integer. A second tooth portion has a first side defined by a third slot and a second side defined by a fourth slot adjacent to the third slot. The second tooth portion has a second width that is the photovoltaic cell width multiplied by a second positive integer different than the first positive integer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1981467	12/1933	Radtke	N/A	N/A
3156497	12/1963	Lessard	N/A	N/A
3581779	12/1970	Gilbert, Jr.	N/A	N/A
4258948	12/1980	Hoffmann	N/A	N/A
4349220	12/1981	Carroll et al.	N/A	N/A
4499702	12/1984	Turner	N/A	N/A
4636577	12/1986	Peterpaul	N/A	N/A
5167579	12/1991	Rotter	N/A	N/A
5437735	12/1994	Younan et al.	N/A	N/A
5590495	12/1996	Bressler et al.	N/A	N/A
5642596	12/1996	Waddington	N/A	N/A
5992112	12/1998	Josey	N/A	N/A
6008450	12/1998	Ohtsuka et al.	N/A	N/A
6033270	12/1999	Stuart	N/A	N/A
6046399	12/1999	Kapner	N/A	N/A
6105331	12/1999	Rinklake et al.	N/A	N/A
6176059	12/2000	Cantarano et al.	N/A	N/A
6201180	12/2000	Meyer et al.	N/A	N/A
6220329	12/2000	King et al.	N/A	N/A
6308482	12/2000	Strait	N/A	N/A
6320114	12/2000	Kuechler	N/A	N/A
6320115	12/2000	Kataoka et al.	N/A	N/A
6336304	12/2001	Mimura et al.	N/A	N/A
6341454	12/2001	Koleoglou	N/A	N/A
6407329	12/2001	Iino et al.	N/A	N/A
6550215	12/2002	Pulte	N/A	N/A
6576830	12/2002	Nagao et al.	N/A	N/A
6928781	12/2004	Desbois et al.	N/A	N/A
6972367	12/2004	Federspiel et al.	N/A	N/A
7138578	12/2005	Komamine	N/A	N/A

7155870	12/2006	Almy	N/A	N/A
7178295	12/2006	Dinwoodie	N/A	N/A
7487771	12/2008	Eiffert et al.	N/A	N/A
7587864	12/2008	McCaskill et al.	N/A	N/A
7666491	12/2009	Yang et al.	N/A	N/A
7678990	12/2009	McCaskill et al.	N/A	N/A
7678991	12/2009	McCaskill et al.	N/A	N/A
7748191	12/2009	Podirsky	N/A	N/A
7819114	12/2009	Augenbraun et al.	N/A	N/A
7824191	12/2009	Podirsky	N/A	N/A
7832176	12/2009	McCaskill et al.	N/A	N/A
8118109	12/2011	Hacker	N/A	N/A
8168880	12/2011	Jacobs et al.	N/A	N/A
8173889	12/2011	Kalkanoglu et al.	N/A	N/A
8210570	12/2011	Railkar et al.	N/A	N/A
8276329	12/2011	Lenox	N/A	N/A
8312693	12/2011	Cappelli	N/A	N/A
8319093	12/2011	Kalkanoglu et al.	N/A	N/A
8333040	12/2011	Shiao et al.	N/A	N/A
8371076	12/2012	Jones et al.	N/A	N/A
8375653	12/2012	Shiao et al.	N/A	N/A
8404967	12/2012	Kalkanoglu et al.	N/A	N/A
8410349	12/2012	Kalkanoglu et al.	N/A	N/A
8418415	12/2012	Shiao et al.	N/A	N/A
8438796	12/2012	Shiao et al.	N/A	N/A
8468754	12/2012	Railkar et al.	N/A	N/A
8468757	12/2012	Krause et al.	N/A	N/A
8505249	12/2012	Geary	N/A	N/A
8512866	12/2012	Taylor	N/A	N/A
8513517	12/2012	Kalkanoglu et al.	N/A	N/A
8586856	12/2012	Kalkanoglu et al.	N/A	N/A
8601754	12/2012	Jenkins et al.	N/A	N/A
8623499	12/2013	Viasnoff	N/A	N/A
8629578	12/2013	Kurs et al.	N/A	N/A
8646228	12/2013	Jenkins	N/A	N/A
8656657	12/2013	Livsey et al.	N/A	N/A
8671630	12/2013	Lena et al.	N/A	N/A
8677702	12/2013	Jenkins	N/A	N/A
8695289	12/2013	Koch et al.	N/A	N/A
8713858	12/2013	Xie	N/A	N/A
8713860	12/2013	Railkar et al.	N/A	N/A
8733036	12/2013	Salam	N/A	N/A
8733038	12/2013	Kalkanoglu et al.	N/A	N/A
8776455	12/2013	Azoulay	N/A	N/A
8789321	12/2013	Ishida	N/A	N/A
8793940	12/2013	Kalkanoglu et al.	N/A	N/A
8793941	12/2013	Bosler et al.	N/A	N/A
8826607	12/2013	Shiao et al.	N/A	N/A
8835751	12/2013	Kalkanoglu et al.	N/A	N/A
8863451	12/2013	Jenkins et al.	N/A	N/A

8898970	12/2013	Jenkins et al.	N/A	N/A
8925262	12/2014	Railkar et al.	N/A	N/A
8943766	12/2014	Gombarick et al.	N/A	N/A
8946544	12/2014	Jabos et al.	N/A	N/A
8950128	12/2014	Kalkanoglu et al.	N/A	N/A
8959848	12/2014	Jenkins et al.	N/A	N/A
8966838	12/2014	Jenkins	N/A	N/A
8966850	12/2014	Jenkins et al.	N/A	N/A
8994224	12/2014	Mehta et al.	N/A	N/A
9032672	12/2014	Livsey et al.	N/A	N/A
9059348	12/2014	Quinlan et al.	N/A	N/A
9145498	12/2014	Ultsch	N/A	N/A
9153950	12/2014	Yamanaka et al.	N/A	N/A
9166087	12/2014	Chihlas et al.	N/A	N/A
9169646	12/2014	Rodrigues et al.	N/A	N/A
9170034	12/2014	Bosler et al.	N/A	N/A
9171991	12/2014	Pearce	N/A	N/A
9178465	12/2014	Shiao et al.	N/A	N/A
9202955	12/2014	Livsey et al.	N/A	N/A
9212832	12/2014	Jenkins	N/A	N/A
9217584	12/2014	Kalkanoglu et al.	N/A	N/A
9270221	12/2015	Zhao	N/A	N/A
9273885	12/2015	Rordigues et al.	N/A	N/A
9276141	12/2015	Kalkanoglu et al.	N/A	N/A
9331224	12/2015	Koch et al.	N/A	N/A
9356174	12/2015	Duarte et al.	N/A	N/A
9359014	12/2015	Yang et al.	N/A	N/A
9412890	12/2015	Meyers	N/A	N/A
9528270	12/2015	Jenkins et al.	N/A	N/A
9605432	12/2016	Robbins	N/A	N/A
9670353	12/2016	Peng et al.	N/A	N/A
9711672	12/2016	Wang	N/A	N/A
9755573	12/2016	Livsey et al.	N/A	N/A
9786802	12/2016	Shiao et al.	N/A	N/A
9831818	12/2016	West	N/A	N/A
D805211	12/2016	Wu	N/A	N/A
9912284	12/2017	Svec	N/A	N/A
9920515	12/2017	Xing et al.	N/A	N/A
9923515	12/2017	Rodrigues et al.	N/A	N/A
9938729	12/2017	Coon	N/A	N/A
9987786	12/2017	Stoiljkovic et al.	N/A	N/A
9991412	12/2017	Gonzalez et al.	N/A	N/A
9998067	12/2017	Kalkanoglu et al.	N/A	N/A
10015933	12/2017	Boldrin	N/A	N/A
10027273	12/2017	West et al.	N/A	N/A
10115850	12/2017	Rodrigues et al.	N/A	N/A
10128660	12/2017	Apte et al.	N/A	N/A
10156075	12/2017	McDonough	N/A	N/A
10179852	12/2018	Gossi et al.	N/A	N/A
10187005	12/2018	Rodrigues et al.	N/A	N/A

10256765	12/2018	Rodrigues et al.	N/A	N/A
10284136	12/2018	Mayfield et al.	N/A	N/A
10454408	12/2018	Livsey et al.	N/A	N/A
10480192	12/2018	Xing et al.	N/A	N/A
10530292	12/2019	Cropper et al.	N/A	N/A
10560048	12/2019	Fisher et al.	N/A	N/A
10563406	12/2019	Kalkanoglu et al.	N/A	N/A
D879031	12/2019	Lance et al.	N/A	N/A
10579028	12/2019	Jacob	N/A	N/A
10584493	12/2019	Jenkins et al.	N/A	N/A
10669414	12/2019	Li et al.	N/A	N/A
10738473	12/2019	De Ridder et al.	N/A	N/A
10784813	12/2019	Kalkanoglu et al.	N/A	N/A
D904289	12/2019	Lance et al.	N/A	N/A
10907355	12/2020	Hubbard et al.	N/A	N/A
10914063	12/2020	Lee et al.	N/A	N/A
RE48555	12/2020	Cancio et al.	N/A	N/A
11012025	12/2020	Seery	N/A	H02S 30/10
11012026	12/2020	Kalkanoglu et al.	N/A	N/A
11015085	12/2020	Bruns et al.	N/A	N/A
11065849	12/2020	Ackermann et al.	N/A	N/A
11177639	12/2020	Nguyen et al.	N/A	N/A
11217715	12/2021	Sharenko et al.	N/A	N/A
11251744	12/2021	Bunea	N/A	N/A
11258399	12/2021	Kalkanoglu et al.	N/A	N/A
11283394	12/2021	Perkins et al.	N/A	N/A
11309828	12/2021	Sirski et al.	N/A	N/A
11394344	12/2021	Perkins et al.	N/A	N/A
11424379	12/2021	Sharenko et al.	N/A	N/A
11431280	12/2021	Liu et al.	N/A	N/A
11431281	12/2021	Perkins et al.	N/A	N/A
11444569	12/2021	Clemente et al.	N/A	N/A
11454027	12/2021	Kuiper et al.	N/A	N/A
11459757	12/2021	Nguyen et al.	N/A	N/A
11486144	12/2021	Bunea et al.	N/A	N/A
11489482	12/2021	Peterson et al.	N/A	N/A
11496088	12/2021	Sirski et al.	N/A	N/A
11508861	12/2021	Perkins et al.	N/A	N/A
11512480	12/2021	Achor et al.	N/A	N/A
11527665	12/2021	Boitnott	N/A	N/A
11545927	12/2022	Abra et al.	N/A	N/A
11545928	12/2022	Perkins et al.	N/A	N/A
11658470	12/2022	Nguyen et al.	N/A	N/A
11661745	12/2022	Bunea et al.	N/A	N/A
11689149	12/2022	Clemente et al.	N/A	N/A
11705531	12/2022	Sharenko et al.	N/A	N/A
11728759	12/2022	Nguyen et al.	N/A	N/A
11732490	12/2022	Achor et al.	N/A	N/A
11811361	12/2022	Farhangi et al.	N/A	N/A
11824486	12/2022	Nguyen et al.	N/A	N/A

11824487	12/2022	Nguyen et al.	N/A	N/A
11843067	12/2022	Nguyen et al.	N/A	N/A
2002/0053360	12/2001	Kinoshita et al.	N/A	N/A
2002/0102422	12/2001	Hubbard et al.	N/A	N/A
2002/0129849	12/2001	Heckerorth	N/A	N/A
2003/0101662	12/2002	Ullman	N/A	N/A
2003/0132265	12/2002	Villela et al.	N/A	N/A
2003/0217768	12/2002	Guha	N/A	N/A
2004/0000334	12/2003	Ressler	N/A	N/A
2004/0221886	12/2003	Oono	N/A	N/A
2005/0030187	12/2004	Peress et al.	N/A	N/A
2005/0115603	12/2004	Yoshida et al.	N/A	N/A
2005/0144870	12/2004	Dinwoodie	N/A	N/A
2005/0178428	12/2004	Laaly et al.	N/A	N/A
2005/0193673	12/2004	Rodrigues et al.	N/A	N/A
2006/0042683	12/2005	Gangemi	N/A	N/A
2006/0046084	12/2005	Yang et al.	N/A	N/A
2007/0074757	12/2006	Mellott et al.	N/A	N/A
2007/0181174	12/2006	Ressler	N/A	N/A
2007/0193618	12/2006	Bressler et al.	N/A	N/A
2007/0249194	12/2006	Liao	N/A	N/A
2007/0295385	12/2006	Sheats et al.	N/A	N/A
2008/0006323	12/2007	Kalkanoglu et al.	N/A	N/A
2008/0035140	12/2007	Placer et al.	N/A	N/A
2008/0078440	12/2007	Lim et al.	N/A	N/A
2008/0185748	12/2007	Kalkanoglu	N/A	N/A
2008/0271774	12/2007	Kalkanoglu et al.	N/A	N/A
2008/0302030	12/2007	Stancel et al.	N/A	N/A
2008/0315061	12/2007	Fath	N/A	N/A
2009/0000222	12/2008	Kalkanoglu et al.	N/A	N/A
2009/0014057	12/2008	Croft et al.	N/A	N/A
2009/0014058	12/2008	Croft et al.	N/A	N/A
2009/0019795	12/2008	Szacsvey et al.	N/A	N/A
2009/0044850	12/2008	Kimberley	N/A	N/A
2009/0114261	12/2008	Stancel et al.	N/A	N/A
2009/0133340	12/2008	Shiao et al.	N/A	N/A
2009/0159118	12/2008	Kalkanoglu et al.	N/A	N/A
2009/0178350	12/2008	Kalkanoglu et al.	N/A	N/A
2009/0229652	12/2008	Mapel et al.	N/A	N/A
2009/0275247	12/2008	Richter et al.	N/A	N/A
2010/0019580	12/2009	Croft et al.	N/A	N/A
2010/0095618	12/2009	Edison et al.	N/A	N/A
2010/0101634	12/2009	Frank et al.	N/A	N/A
2010/0116325	12/2009	Nikoonahad	N/A	N/A
2010/0131108	12/2009	Meyer	N/A	N/A
2010/0139184	12/2009	Williams et al.	N/A	N/A
2010/0146878	12/2009	Koch et al.	N/A	N/A
2010/0159221	12/2009	Kourtakis et al.	N/A	N/A
2010/0170169	12/2009	Railkar et al.	N/A	N/A
2010/0186798	12/2009	Tormen et al.	N/A	N/A

2010/0242381	12/2009	Jenkins	N/A	N/A
2010/0313499	12/2009	Gangemi	N/A	N/A
2010/0325976	12/2009	DeGenfelder et al.	N/A	N/A
2010/0326488	12/2009	Aue et al.	N/A	N/A
2010/0326501	12/2009	Zhao et al.	N/A	N/A
2011/0030761	12/2010	Kalkanoglu et al.	N/A	N/A
2011/0036386	12/2010	Browder	N/A	N/A
2011/0036389	12/2010	Hardikar et al.	N/A	N/A
2011/0048507	12/2010	Livsey et al.	N/A	N/A
2011/0058337	12/2010	Han et al.	N/A	N/A
2011/0061326	12/2010	Jenkins	N/A	N/A
2011/0100436	12/2010	Cleereman et al.	N/A	N/A
2011/0104488	12/2010	Muessig et al.	N/A	N/A
2011/0132427	12/2010	Kalkanoglu et al.	N/A	N/A
2011/0168238	12/2010	Metin et al.	N/A	N/A
2011/0239555	12/2010	Cook et al.	N/A	N/A
2011/0302859	12/2010	Crasnianski	N/A	N/A
2011/0314753	12/2010	Farmer et al.	N/A	N/A
2012/0034799	12/2011	Hunt	N/A	N/A
2012/0060434	12/2011	Jacobs	N/A	N/A
2012/0060902	12/2011	Drake	N/A	N/A
2012/0085392	12/2011	Albert et al.	N/A	N/A
2012/0137600	12/2011	Jenkins	N/A	N/A
2012/0176077	12/2011	Oh et al.	N/A	N/A
2012/0212065	12/2011	Cheng et al.	N/A	N/A
2012/0233940	12/2011	Perkins et al.	N/A	N/A
2012/0240490	12/2011	Gangemi	N/A	N/A
2012/0260977	12/2011	Stancel	N/A	N/A
2012/0266942	12/2011	Komatsu et al.	N/A	N/A
2012/0279150	12/2011	Pisllkak et al.	N/A	N/A
2012/0282437	12/2011	Clark et al.	N/A	N/A
2012/0291848	12/2011	Sherman et al.	N/A	N/A
2013/0008499	12/2012	Verger et al.	N/A	N/A
2013/0014455	12/2012	Grieco	N/A	N/A
2013/0118558	12/2012	Sherman	N/A	N/A
2013/0193769	12/2012	Mehta et al.	N/A	N/A
2013/0247988	12/2012	Reese et al.	N/A	N/A
2013/0284267	12/2012	Plug et al.	N/A	N/A
2013/0306137	12/2012	Ko	N/A	N/A
2014/0090697	12/2013	Rodrigues et al.	N/A	N/A
2014/0150843	12/2013	Pearce et al.	N/A	N/A
2014/0173997	12/2013	Jenkins	N/A	N/A
2014/0179220	12/2013	Railkar et al.	N/A	N/A
2014/0182222	12/2013	Kalkanoglu et al.	N/A	N/A
2014/0208675	12/2013	Beerer et al.	N/A	N/A
2014/0254776	12/2013	O'Connor et al.	N/A	N/A
2014/0266289	12/2013	Della Sera et al.	N/A	N/A
2014/0311556	12/2013	Feng et al.	N/A	N/A
2014/0352760	12/2013	Haynes et al.	N/A	N/A
2014/0366464	12/2013	Rodrigues et al.	N/A	N/A

2015/0024159	12/2014	Bess et al.	N/A	N/A
2015/0089895	12/2014	Leitch	N/A	N/A
2015/0162459	12/2014	Lu et al.	N/A	N/A
2015/0340516	12/2014	Kim et al.	N/A	N/A
2015/0349173	12/2014	Morad et al.	N/A	N/A
2015/0354224	12/2014	Maurer et al.	N/A	N/A
2016/0105144	12/2015	Haynes et al.	N/A	N/A
2016/0142008	12/2015	Lopez et al.	N/A	N/A
2016/0254776	12/2015	Rodrigues et al.	N/A	N/A
2016/0276508	12/2015	Huang et al.	N/A	N/A
2016/0359451	12/2015	Mao et al.	N/A	N/A
2017/0159292	12/2016	Chihlas et al.	N/A	N/A
2017/0179319	12/2016	Yamashita et al.	N/A	N/A
2017/0179726	12/2016	Garrity et al.	N/A	N/A
2017/0203555	12/2016	Wang et al.	N/A	N/A
2017/0237390	12/2016	Hudson et al.	N/A	N/A
2017/0331415	12/2016	Koppi et al.	N/A	N/A
2018/0094438	12/2017	Wu et al.	N/A	N/A
2018/0094439	12/2017	Wang et al.	N/A	N/A
2018/0097472	12/2017	Anderson et al.	N/A	N/A
2018/0115275	12/2017	Flanigan et al.	N/A	N/A
2018/0254738	12/2017	Yang et al.	N/A	N/A
2018/0281347	12/2017	Gossi	N/A	N/A
2018/0294765	12/2017	Friedrich et al.	N/A	N/A
2018/0351502	12/2017	Almy et al.	N/A	N/A
2018/0367089	12/2017	Stutterheim et al.	N/A	N/A
2019/0030867	12/2018	Sun et al.	N/A	N/A
2019/0081436	12/2018	Onodi et al.	N/A	N/A
2019/0123679	12/2018	Rodrigues et al.	N/A	N/A
2019/0253022	12/2018	Hardar et al.	N/A	N/A
2019/0305717	12/2018	Allen et al.	N/A	N/A
2020/0020819	12/2019	Farhangi	N/A	H01L 31/042
2020/0109320	12/2019	Jiang	N/A	N/A
2020/0144958	12/2019	Rodrigues et al.	N/A	N/A
2020/0220819	12/2019	Vu et al.	N/A	N/A
2020/0224419	12/2019	Boss et al.	N/A	N/A
2020/0343397	12/2019	Hem-Jensen	N/A	N/A
2021/0002898	12/2020	Knebel et al.	N/A	N/A
2021/0083619	12/2020	Hegedus	N/A	N/A
2021/0095474	12/2020	Yang et al.	N/A	N/A
2021/0113970	12/2020	Stainer et al.	N/A	N/A
2021/0115223	12/2020	Bonekamp et al.	N/A	N/A
2021/0159353	12/2020	Li et al.	N/A	N/A
2021/0171808	12/2020	Ackermann et al.	N/A	N/A
2021/0172174	12/2020	Ackermann et al.	N/A	N/A
2021/0301536	12/2020	Baggs et al.	N/A	N/A
2021/0343886	12/2020	Sharenko et al.	N/A	N/A
2022/0149213	12/2021	Mensink et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2829440	12/2018	CA	N/A
700095	12/2009	CH	N/A
202797032	12/2012	CN	N/A
217150978	12/2021	CN	N/A
1958248	12/1970	DE	N/A
1039361	12/1999	EP	N/A
1837162	12/2006	EP	N/A
1774372	12/2010	EP	N/A
2446481	12/2011	EP	N/A
2784241	12/2013	EP	N/A
3772175	12/2020	EP	N/A
10046767	12/1997	JP	N/A
2002-106151	12/2001	JP	N/A
2001-098703	12/2001	JP	N/A
2017-027735	12/2016	JP	N/A
2018053707	12/2017	JP	N/A
20090084060	12/2008	KR	N/A
10-1348283	12/2013	KR	N/A
10-2019-0000367	12/2018	KR	N/A
10-2253483	12/2020	KR	N/A
2026856	12/2021	NL	N/A
2010/151777	12/2009	WO	N/A
2011/049944	12/2010	WO	N/A
2015/133632	12/2014	WO	N/A
2018/000589	12/2017	WO	N/A
2019/201416	12/2018	WO	N/A
2020-159358	12/2019	WO	N/A
2021-247098	12/2020	WO	N/A

OTHER PUBLICATIONS

Sunflare, Procduts: “Sunflare Develops Prototype For New Residential Solar Shingles”; 2019 <<sunflaresolar.com/news/sunflare-develops-prototype-for-new-residential-solar-shingles>>

retrieved Feb. 2, 2021. cited by applicant

RGS Energy, 3.5kW PowerHouse 3.0 system installed in an afternoon; Jun. 7, 2019

<<facebook.com/RGSEnergy/>> retrieved Feb. 2, 2021. cited by applicant

Tesla, Solar Roof <<tesla.com/solarroof>> retrieved Feb. 2, 2021. cited by applicant

“Types of Roofing Underlayment”, Owens Corning Roofing;

<<<https://www.owenscorning.com/en-us/roofing/tools/how-roofing-underlayment-helps-protect-your-home>>> retrieved Nov. 1, 2021. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/499,516, filed on Oct. 12, 2021, entitled “SOLAR ROOFING SYSTEM”, which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/091,017, filed Oct. 13, 2020, entitled “SOLAR ROOFING SYSTEM,” the contents of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to solar roofing systems including roof-integrated photovoltaic modules and roofing shingles. More particularly, the present invention relates to solar roofing systems including roof-integrated photovoltaic modules and roofing shingles having elements with matching widths.

BACKGROUND

(2) Solar modules can be placed on building roofs (e.g., residential roofs) to generate electricity. One obstacle to mass-market adoption of solar roofing is poor aesthetics. Standard rack-mounted photovoltaic (“PV”) systems have a very different appearance than traditional roofing materials (e.g., asphalt shingles, wooden shingles, slate shingles, etc.), which can draw unwanted attention. Even low-profile PV systems still receive poor aesthetic feedback from consumers.

(3) Specifically, typical PV module materials and circuit formations include PV elements having a constant width and a grid-like appearance, while typical roofing shingles include elements having irregular viewed widths, causing the PV modules not to visually match the look of standard roofing shingles.

SUMMARY

(4) In some embodiments, a system, comprising a photovoltaic module comprising a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has a photovoltaic cell width; and a roofing shingle having a top surface and a bottom surface, the roofing shingle having an exposure zone at a lower end of the top surface and a headlap zone at an upper end of the top surface, wherein a plurality of slots extends from the lower end toward the headlap zone, wherein the plurality of slots defines a plurality of tooth portions therebetween, wherein a first one of the plurality of tooth portions has a first side that is defined by a first one of the plurality of slots and a second side that is defined by a second one of the plurality of slots that is adjacent to the first one of the plurality of slots, wherein the first one of the plurality of tooth portions has a first width that is measured from the first one of the plurality of slots to the second one of the plurality of slots, wherein the first width is the photovoltaic cell width multiplied by a first positive integer, wherein a second one of the plurality of tooth portions has a first side that is defined by a third one of the plurality of slots and a second side that is defined by a fourth one of the plurality of slots that is adjacent to the third one of the plurality of slots, wherein the second one of the plurality of tooth portions has a second width that is measured from the third one of the plurality of slots to the fourth one of the plurality of slots, and wherein the second width is the photovoltaic cell width multiplied by a second positive integer that is different than the first positive integer.

(5) In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by a sixth one of the plurality of slots that is adjacent to the fifth one of the plurality of slots, and wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the sixth one of the plurality of slots, wherein the third width is the photovoltaic cell width multiplied by 0.5 and by a third positive integer that is different than the first positive integer and different than the second positive integer. In some embodiments, each of the first and second positive integers is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,

13, 14, 15, 16, 17, 18, 19, and 20. In some embodiments, the roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt. In some embodiments, the top surface of the roofing shingle comprises embedded granules.

(6) In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by one of either (a) the first side, or (b) the second side, and wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the one of either the first side or the second side, wherein the third width is the photovoltaic cell width multiplied by 0.5 and by a third positive integer that is different than the first positive integer and different than the second positive integer.

(7) In some embodiments, the system further includes a second roofing shingle having a top surface and a bottom surface, the second roofing shingle having an exposure zone at a lower end of the top surface of the second roofing shingle and a headlap zone at an upper end of the top surface of the second roofing shingle, wherein a plurality of slots extends from the lower end of the second roofing shingle toward the headlap zone of the second roofing shingle, wherein the plurality of slots of the second roofing shingle defines a plurality of tooth portions therebetween, wherein a first one of the plurality of tooth portions of the second roofing shingle has a first side that is defined by a first one of the plurality of slots of the second roofing shingle and a second side that is defined by a second one of the plurality of slots of the second roofing shingle that is adjacent to the first one of the plurality of slots of the second roofing shingle, wherein the first one of the plurality of tooth portions of the second roofing shingle has a third width that is measured from the first one of the plurality of slots of the second roofing shingle to the second one of the plurality of slots of the second roofing shingle, wherein the third width is the photovoltaic cell width multiplied by a third positive integer, wherein a second one of the plurality of tooth portions of the second roofing shingle has a first side that is defined by a third one of the plurality of slots of the second roofing shingle and a second side that is defined by a fourth one of the plurality of slots of the second roofing shingle that is adjacent to the third one of the plurality of slots of the second roofing shingle, wherein the second one of the plurality of tooth portions of the second roofing shingle has a fourth width that is measured from the third one of the plurality of slots of the second roofing shingle to the fourth one of the plurality of slots of the second roofing shingle, and wherein the fourth width is the photovoltaic cell width multiplied by a fourth positive integer that is different than the third positive integer.

(8) In some embodiments, an arrangement of the tooth portions of the second roofing shingle is not identical to an arrangement of the tooth portions of the roofing shingle. In some embodiments, an arrangement of the tooth portions of the second roofing shingle is identical to an arrangement of the tooth portions of the roofing shingle. In some embodiments, the third one of the plurality of slots is a same one of the plurality of slots as the second one of the plurality of slots, and wherein the first one of the plurality of tooth portions is adjacent to the second one of the plurality of tooth portions.

(9) In some embodiments, the system further includes a wireway configured to be positioned between the photovoltaic module and a further photovoltaic module that is adjacent to the photovoltaic module, wherein the wireway is configured to enclose at least one electrical cable, wherein a width of the wireway as measured in a horizontal direction between the photovoltaic module and the further photovoltaic module is the photovoltaic cell width multiplied by two, wherein the wireway includes a dark colored portion and a light colored portion, and wherein the light colored portion extends across the wireway in a vertical direction that is perpendicular to the horizontal direction. In some embodiments, the light-colored portion is positioned at an edge of the wireway that is adjacent to the photovoltaic module. In some embodiments, the light-colored portion is positioned halfway intermediate (1) an edge of the wireway that is adjacent to the photovoltaic module and (2) an edge of the wireway that is adjacent to the further photovoltaic module. In some embodiments, the wireway further comprises a further light colored portion

extending across a bottom edge of the wireway in the horizontal direction.

(10) In some embodiments, a roofing shingle includes a top surface and a bottom surface, the roofing shingle having an exposure zone at a lower end of the top surface and a headlap zone at an upper end of the top surface, wherein the roofing shingle is configured to be installed on a roof adjacent to a photovoltaic module including a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has a photovoltaic cell width, wherein a plurality of slots extends from the lower end toward the headlap zone, wherein the plurality of slots defines a plurality of tooth portions therebetween, wherein a first one of the plurality of tooth portions has a first side that is defined by a first one of the plurality of slots and a second side that is defined by a second one of the plurality of slots that is adjacent to the first one of the plurality of slots, wherein the first one of the plurality of tooth portions has a first width that is measured from the first one of the plurality of slots to the second one of the plurality of slots, wherein the first width is the photovoltaic cell width multiplied by a first positive integer, wherein a second one of the plurality of tooth portions has a first side that is defined by a third one of the plurality of slots and a second side that is defined by a fourth one of the plurality of slots that is adjacent to the third one of the plurality of slots, wherein the second one of the plurality of tooth portions has a second width that is measured from the third one of the plurality of slots to the fourth one of the plurality of slots, and wherein the second width is the photovoltaic cell width multiplied by a second positive integer that is different than the first positive integer.

(11) In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by a sixth one of the plurality of slots that is adjacent to the fifth one of the plurality of slots, wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the sixth one of the plurality of slots, and wherein the third width is the photovoltaic cell width multiplied by a third positive integer that is different than the first positive integer and different than the second positive integer. In some embodiments, each of the first and second positive integers is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20. In some embodiments, the roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt. In some embodiments, the top surface of the roofing shingle comprises embedded granules. In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by one of either (a) the first side, or (b) the second side, and wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the one of either the first side or the second side, wherein the third width is the photovoltaic cell width multiplied by a third positive integer that is different than the first positive integer and different than the second positive integer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a perspective view of an exemplary PV module.
- (2) FIG. 2A shows a schematic view of elements of a layered structure of an exemplary PV module before lamination.
- (3) FIG. 2B shows a schematic view of a layered structure of an exemplary PV module formed by lamination of the elements shown in FIG. 2A.
- (4) FIG. 3 shows an exemplary roofing shingle.
- (5) FIG. 4A shows a first variant of an exemplary roofing shingle.
- (6) FIG. 4B shows a second variant of an exemplary roofing shingle.
- (7) FIG. 4C shows a third variant of an exemplary roofing shingle.

- (8) FIG. 4D shows a fourth variant of an exemplary roofing shingle.
- (9) FIG. 5 shows exemplary embodiments of a PV module.
- (10) FIG. 6A shows an exemplary embodiment of a wireway.
- (11) FIG. 6B shows an exemplary embodiment of a wireway.
- (12) FIG. 6C shows an exemplary embodiment of a wireway.
- (13) FIGS. 6D through 6F show another exemplary embodiment of a wireway.
- (14) FIG. 7 shows an exemplary embodiment of a masking element.
- (15) FIG. 8A shows an exemplary embodiment of a roofing system.
- (16) FIG. 8B shows a magnified view of a first portion of the roofing system of FIG. 8A.
- (17) FIG. 8C shows a magnified view of a second portion of the roofing system of FIG. 8A.
- (18) FIG. 8D shows a magnified view of a third portion of the roofing system of FIG. 8A.
- (19) FIG. 8E shows a magnified view of a fourth portion of the roofing system of FIG. 8A.

DETAILED DESCRIPTION

(20) The present invention will be further explained with reference to the attached drawings, wherein like structures are referred to by like numerals throughout the several views. The drawings shown are not necessarily to scale, with emphasis instead generally being placed upon illustrating the principles of the present invention. Further, some features may be exaggerated to show details of particular components.

(21) The figures constitute a part of this specification and include illustrative embodiments of the present invention and illustrate various objects and features thereof. Further, the figures are not necessarily to scale, some features may be exaggerated to show details of particular components. In addition, any measurements, specifications and the like shown in the figures are intended to be illustrative, and not restrictive. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

(22) Among those benefits and improvements that have been disclosed, other objects and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying figures. Detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely illustrative of the invention that may be embodied in various forms. In addition, each of the examples given in connection with the various embodiments of the invention which are intended to be illustrative, and not restrictive.

(23) Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases “in one embodiment” and “in some embodiments” as used herein do not necessarily refer to the same embodiment(s), though they may. Furthermore, the phrases “in another embodiment” and “in some other embodiments” as used herein do not necessarily refer to a different embodiment, although they may. Thus, as described below, various embodiments of the invention may be readily combined, without departing from the scope or spirit of the invention.

(24) The term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.”

(25) The exemplary embodiments relate to a roofing system having elements (e.g., roofing shingles, PV modules, wireways, and masking elements) having visual elements the width of which is harmonized around a constant base width. In some embodiments, the constant base width is a cell pitch. In some embodiments, the cell pitch is defined as the width of a PV element plus the width of a space that exists between two adjacent PV elements. In some embodiments, such harmonization provides a consistent and blended aesthetic appearance across such a roofing system, as will be discussed in further detail hereinafter.

(26) In some embodiments, a solar roofing system includes at least one PV module and at least one roofing shingle. In some embodiments, each of the at least one PV modules includes a plurality of PV cells. In some embodiments, each of the PV cells has a PV cell width that is the same for all of the PV cells. In some embodiments, the cell width is a “half-cut” width, e.g., the width of a premanufactured PV cell that has been cut in half.

(27) FIG. 1 shows an exemplary PV module **100**. The exemplary PV module **100** includes a headlap region **110** and a PV region **120**. In some embodiments, the headlap region **110** comprises thermoplastic olefin (“TPO”), polyvinyl chloride (“PVC”), or asphalt. In some embodiments, the headlap region **110** includes embedded granules. In some embodiments, the headlap region **110** defines a nailing line **112** extending across the headlap region **110**. In some embodiments, the nailing line **112** extends across the headlap region **110** approximately midway between the end of the headlap region **110** that borders the PV region and the opposite end of the headlap region **110**. In some embodiments, the nailing line **112** defines an area of the headlap region **110** through which mechanical fasteners (e.g., nails, screws, etc.) can be driven to secure the PV module **100** to a roof deck in the standard manner.

(28) In some embodiments, the PV region **120** includes a plurality of PV portions **122**. In some embodiments, each of the PV portions **122** includes a layered structure that is typical of a laminate PV module, as discussed below with reference to FIGS. 2A and 2B. In some embodiments, the PV region **120** includes grooves **124** separating adjacent ones of the PV portions. In some embodiments, each of the PV portions **122** is separately formed from others of the PV portions **122**, and the grooves **124** are formed by spaces between adjacent ones of the PV portions **122**. In some embodiments, the PV portions **122** forming the PV region **120** are integrally formed with one another (e.g., form a single layered structure) and the grooves **124** are formed in a superstrate layer thereof. In some embodiments, the grooves **124** between adjacent ones of the PV portions **122** provide the appearance of discrete portions similar to those of conventional shingles. In some embodiments, the PV region **120** is formed atop material of the headlap region **110** (e.g., the substrate of the PV region **120** is deposited on the material of the headlap region **110**). In some embodiments, the PV region **120** and the headlap region **110** join one another end-to-end.

(29) In some embodiments, the PV module **100** includes a junction box **130**. In some embodiments, the junction box **130** is positioned at an end of the headlap region **110** that is opposite the PV region **120**. In some embodiments, the junction box **130** is positioned at a center of an end of the headlap region **110** that is opposite the PV region **120**. In some embodiments, the junction box **130** is electrically connected to the PV region **120** by electrical connectors (e.g., wires) that traverse (e.g., pass under or through) the headlap region **110**.

(30) In some embodiments, such as the PV module **100** discussed above with reference to FIG. 1, an exemplary PV module includes a layered structure. FIGS. 2A and 2B show an exemplary embodiment of a layered structure **200** that, in some embodiments, forms part of an exemplary PV module.

(31) FIG. 2A shows an exploded view of the layers of the layered structure **200** prior to lamination to form the layered structure **200**. FIG. 2B shows the layered structure following lamination. It will be apparent to those of skill in the art that FIGS. 2A and 2B present schematic views of the layered structure **200** and are not intended to provide a to-scale representation.

(32) Referring now to FIG. 2A, in some embodiments, the layered structure **200** includes a superstrate layer **210** that forms an upper surface of the layered structure **200** and of the PV module **100** (i.e., the surface that, when the PV module **100** is installed on a roof, faces away from the roof and toward the sun). In some embodiments, the superstrate layer **210** has an upper surface **212** (i.e., the side of the superstrate layer **210** that faces toward the sun when installed as described above) and a lower surface **214** opposite the upper surface **212**. In some embodiments, the superstrate layer **210** is optically transparent (e.g., it has a solar weighted transmittance of 80% or greater). In some embodiments, the superstrate provides electrical insulation and moisture resistance. In some

embodiments, the superstrate layer **210** comprises a glass material, such as low-iron solar glass. In some embodiments, the superstrate layer **210** comprises a polymeric material such as ethylene tetrafluoroethylene (“ETFE”), polyethylene terephthalate (“PET”), or an acrylic such as polymethyl methacrylate (“PMMIA”). In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 250 microns. In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 200 microns. In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 150 microns. In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 100 microns. In some embodiments, the superstrate layer **210** has a thickness of from 100 microns to 250 microns. In some embodiments, the superstrate layer **210** has a thickness of from 100 microns to 200 microns. In some embodiments, the superstrate layer **210** has a thickness of from 100 microns to 150 microns. In some embodiments, the superstrate layer **210** has a thickness of from 150 microns to 250 microns. In some embodiments, the superstrate layer **210** has a thickness of from 150 microns to 200 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 250 microns.

(33) In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 350 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 300 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 350 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 300 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 350 microns. In some embodiments, the superstrate layer **210** has a thickness of from 350 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 350 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 350 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 400 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 400 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 450 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 325 microns to 375 microns. In some embodiments, the superstrate layer **210** has a thickness of about 300 microns. In some embodiments, the superstrate layer **210** has a thickness of 300 microns.

(34) In some embodiments, the superstrate layer **210** has a thickness of from 1.6 millimeters to 4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 1.6 millimeters to 3.2 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 1.6 millimeters to 2.4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 2.4 millimeters to 4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 2.4 millimeters to 3.2 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 3.2 millimeters to 4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 2.8 millimeters to 3.6 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 3 millimeters to 3.4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 3.1 millimeters to 3.3

millimeters. In some embodiments, the superstrate layer **210** has a thickness about 3.2 millimeters. In some embodiments, the superstrate layer **210** has a thickness of 3.2 millimeters.

(35) Continuing to refer to FIG. **2A**, in some embodiments, the layered structure **200** includes an upper encapsulant layer **220**. In some embodiments, the upper encapsulant layer **220** has an upper surface **222** and a lower surface **224** opposite the upper surface **222**. In some embodiments, the upper surface **222** of the upper encapsulant layer **220** contacts the lower surface **214** of the superstrate layer **210**. In some embodiments, the upper encapsulant layer **220** is optically transparent (e.g., it has a solar weighted transmittance of 80% or greater). In some embodiments, the upper encapsulant layer provides electrical insulation. In some embodiments, the upper encapsulant layer **220** comprises an encapsulating material such as ethylene-co-vinyl acetate (“EVA”), polydimethyl siloxane (“PDMS”), a polyolefin elastomer (“POE”), polyvinyl butyral (“PVB”), polyurethane epoxy, silicone, or an ionomer such as the series of ionomer-based encapsulants commercialized by DuPont de Nemours, Inc. under the trade name PV5400. In some embodiments, the thickness of the upper encapsulant layer **220** varies across the layered structure **200**, as will be discussed in greater detail hereinafter.

(36) Continuing to refer to FIG. **2A**, in some embodiments, the layered structure **200** includes a PV layer **230** having an upper surface **232** and a lower surface **234** opposite the upper surface **232**. In some embodiments, the upper surface **232** of the PV layer **230** contacts the lower surface **224** of the upper encapsulant layer **220**. In some embodiments, the PV layer **230** includes at least one PV element **236** (e.g., at least one PV cell having a constant PV cell width **237** as described above). In some embodiments, the PV layer **230** includes an array of PV elements **236**. In some embodiments in which the PV layer **230** includes a plurality of the PV element **236**, the PV elements **236** are electrically interconnected with one another. In some embodiments, the PV layer **230** includes an array of interconnected PV elements **236**. In some embodiments, gaps are formed between adjacent ones of the PV elements **236**. In some embodiments, the gaps are significantly smaller than the PV elements **236**; for example, in some embodiments, a width of each of the PV elements **236** is 160 millimeters and the gaps are from 2 millimeters to 5 millimeters in size. In some embodiments, the PV layer **230** also includes other active and/or passive electronic components.

(37) Continuing to refer to FIG. **2A**, in some embodiments, the layered structure **200** includes a lower encapsulant layer **240** having an upper surface **242** and a lower surface **244** opposite the upper surface **242**. In some embodiments, the upper surface **242** of the lower encapsulant layer **240** contacts the lower surface **234** of the PV layer **230**. In some embodiments, the lower encapsulant layer **240** provides electrical insulation. In some embodiments, the lower encapsulant layer **240** is optically transparent. In some embodiments, the lower encapsulant layer **240** is not optically transparent. In some embodiments, the thickness of the lower encapsulant layer **240** is in the range of 100 to 1000 microns. In some embodiments, the thickness of the lower encapsulant layer **240** is sufficiently large (e.g., greater than 100 microns) so as to prevent delamination between the PV layer **230** and the substrate **250**. In some embodiments, the thickness of the lower encapsulant layer **240** is consistent across the entirety of the layered structure **200**. In some embodiments, the lower encapsulant layer **240** comprises an encapsulating material such as ethylene-co-vinyl acetate (“EVA”), polydimethyl siloxane (“PDMS”), a polyolefin elastomer (“POE”), polyvinyl butyral (“PVB”), polyurethane epoxy, silicone, or an ionomer such as the series of ionomer-based encapsulants commercialized by DuPont de Nemours, Inc. under the trade name PV5400. In some embodiments, the lower encapsulant layer **240** comprises the same encapsulating material as the upper encapsulant layer **220**.

(38) Continuing to refer to FIG. **2A**, in some embodiments, the layered structure **200** includes a substrate **250** having an upper surface **252** and a lower surface **254** opposite the lower surface **252**. In some embodiments, the upper surface **252** of the substrate **250** contacts the lower surface **244** of the lower encapsulant layer **240**. In some embodiments, the lower surface **254** of the substrate **250** forms the lower surface **204** of the layered structure **200**. In some embodiments, the substrate **250**

provides electrical insulation and moisture resistance. In some embodiments, the substrate **250** is optically transparent. In some embodiments, the substrate **250** is not optically transparent. In some embodiments, the substrate **250** comprises a glass material. In some embodiments, the substrate **250** comprises a polymeric material such as ETFE, PET, an acrylic such as PMMA, polypropylene, polyvinyl chloride (“PVC”), or a glass-reinforced or fiber-reinforced composite such as a material meeting the National Electrical Manufacturers Association (“NEMA”) grades FR-4 or G-10. In some embodiments, the substrate **250** has a thickness in the range of 200 microns to ¼ inch. In some embodiments, the substrate **250** is sufficiently rigid to provide mechanical stiffening to the PV module **100**.

(39) Referring now to FIG. 2B, the layered structure **200** is shown following lamination. In some embodiments, during the lamination process, the encapsulating material of the upper encapsulant layer **220** and the encapsulating material of the lower encapsulant layer **240** are melted and flow within the gaps between adjacent ones of the PV elements **236** shown in FIG. 2A, thereby encapsulating (e.g., surrounding on all sides) each of the PV elements **236** with encapsulating material. In some embodiments, as a result of this process, the PV layer **230** includes encapsulant portions **238** located between adjacent ones of the PV elements **236**, and providing continuity between the encapsulating material of the upper encapsulant layer **220** and the encapsulating material of the lower encapsulant layer **240**. In some embodiments, the resulting region of the layered structure **200** (e.g., the upper encapsulant layer **220**, the PV layer **230**, and the lower encapsulant layer **240**) resembles a single block of encapsulant material with the PV elements positioned therein.

(40) FIG. 3 shows an exemplary roofing shingle **300**. In some embodiments, the roofing shingle **300** comprises thermoplastic olefin (“TPO”), polyvinyl chloride (“PVC”), or asphalt. In some embodiments, roofing shingle **300** includes embedded granules. In some embodiments, the roofing shingle **300** includes a top end **302**, a bottom end **304**, a first side **306**, and a second side. In some embodiments, the roofing shingle **300** includes a headlap region **310** adjacent the top end **302** and a tooth region **320** adjacent the bottom end **304**. In some embodiments, the headlap region **310** defines a nailing line **312** extending across the headlap region **310**. In some embodiments, the nailing line **312** extends across the headlap region **310** approximately midway between the end of the headlap region **310** that borders the tooth region **320** and the opposite end of the headlap region **110**. In some embodiments, the nailing line **312** defines an area of the headlap region **310** through which mechanical fasteners (e.g., nails, screws, etc.) can be driven to secure the roofing shingle **300** to a roof deck in the standard manner.

(41) In some embodiments, the tooth region **320** includes a plurality of slots **322** that are spaced apart along the width of the tooth region **320** from the first side **306** to the second side, and extend from the bottom end **304** toward the top end **302**. In some embodiments, each of the slots **322** has a width that is the same as, or is similar to, the gaps between adjacent ones of the PV elements **236** in the PV module **100**. In some embodiments, the width of each of the slots **322** is ½ inch. In some embodiments, the width of each of the slots **322** is 6 millimeters. In some embodiments, the first side **306**, the slots **322**, and the second side define a plurality of tooth portions **324**, **326**, **328**, **330**, **332**, **334** therebetween. The exemplary roofing shingle **300** shown in FIG. 3 includes six (6) of the tooth portions **324**, **326**, **328**, **330**, **332**, **334**, but it will be apparent to those of skill in the art that various embodiments of the roofing shingle **300** may have any other number of tooth portions. In some embodiments, the width of the tooth portions **324**, **326**, **328**, **330**, **332**, **334** is variable, e.g., for a given instance of the roofing shingle **300**, a first one of the tooth portions (e.g., tooth portion **324**) has a first width, and a second one of the tooth portions (e.g., tooth portion **326**) has a second width that is different than the first width. In some embodiments, each of the tooth portions **324**, **326**, **328**, **330**, **332**, **334** has a width that is an integer multiple of the PV cell width **237** (e.g., is equal to the PV cell width **237**, or is two times, or three times, or four times, or five times, or six times, or seven times, or eight times, or nine times, or ten times, or eleven times, or twelve times,

or thirteen times, or fourteen times, or fifteen times, or sixteen times, or seventeen times, or eighteen times, or nineteen times, or twenty times the PV cell width **237**).

(42) In some embodiments, different ones of the exemplary roofing shingle **300** have differently sized and differently arranged tooth portions. For example, in some embodiments, a manufacturer of the roofing shingle **300** may manufacture different versions of the roofing shingle **300** so as to provide a roofing system including a plurality of the roofing shingle **300** that differ from one another so as to provide a non-uniform appearance to the roofing system. FIGS. **4A-4D** show different variants **410**, **420**, **430**, and **440** of the exemplary roofing shingle **300**. In some embodiments, each of the variants **410**, **420**, **430**, and **440** has a plurality of tooth portions that vary in width among the different tooth portions of any given one of the variants **410**, **420**, **430**, and **440**, and the widths of all tooth portions are integer multiples of the PV cell width **237**. In some embodiments, each of the variants **410**, **420**, **430**, and **440** has tooth portions that differ in arrangement as compared to those of the others of the variants **410**, **420**, **430**, and **440**. In some embodiments, a manufacturer of the roofing shingle **300** may manufacture a suitable number of different variants so as to impart a random appearance to a roofing system incorporating such variants. For example, in some embodiments, a roofing system includes four of the variants **410**, **420**, **430**, and **440**, as shown in FIGS. **4A-4D**, but it will be apparent to those of skill in the art that this is only exemplary, and that any other number of variants may be produced.

(43) In some embodiments, the roofing shingle **300** is composed of a single layer. In some embodiments, the roofing shingle **300** is composed of multiple layers. In some embodiments, the roofing shingle **300** is laminated.

(44) As discussed above, in some embodiments, the assembled PV module **100** includes a plurality of PV elements **236** that are spaced apart by a quantity of encapsulant portions **238** that are positioned between the PV elements **236** as part of the lamination process. In some embodiments, the space between the PV elements **236** that is formed in this manner is referred to as a “cell gap”. In some embodiments, due to uniform sizing and spacing of the PV elements **236**, the PV elements **236** and the cell gaps therebetween provide a uniform, grid-like appearance. In some embodiments, to provide a non-uniform appearance, the cell gap is selectively revealed or hidden.

(45) FIG. **5** shows PV elements **500** forming PV modules **510**, **520**, **530**. For clarity, the PV elements **500** are substantially the only elements of the PV modules **510**, **520**, **530** shown in FIG. **5**, but it will be apparent to those of skill in the art that the PV modules **510**, **520**, **530** will include other elements as described above. Also for clarity, only one of the PV elements **500** forming each of the PV modules **510**, **520**, **530** is specifically called out in FIG. **5**, but it will be apparent to those of skill in the art that discussion of the PV elements **500** may refer to any of the PV elements **500** forming the PV modules **510**, **520**, **530**, whether or not specifically identified in FIG. **5**.

(46) In some embodiments, as shown in FIG. **5**, each of the PV modules **510**, **520**, **530** includes sixteen (16) of the PV elements **500** that are “half-cut” cells having a width of about 90 mm and a cell gap of about 4 mm between adjacent ones of the PV elements, thereby to produce the PV modules **510**, **520**, **530** that are 60 inches wide, but it will be apparent to those of skill in the art that these dimensions are only exemplary. In some embodiments, the PV modules **510**, **520**, **530** include portions **512**, **522**, **532**, respectively, of a color-contrasting material positioned behind the PV elements **500** (e.g., positioned in or on the PV modules **510**, **520**, **530** so as to be positioned between the PV elements and a roof deck to which the PV modules **510**, **520**, **530** are installed). In some embodiments, the color-contrasting material comprises a patterned backsheet. In some embodiments, the color-contrasting material comprises a cloaking tape. As shown in FIG. **5**, the inclusion of the portions **512**, **522**, **532** at different locations within the PV modules **510**, **520**, **530** provides a non-uniform appearance to the PV modules **510**, **520**, **530**, despite each of the PV modules **510**, **520**, **530** having the same arrangement of the PV elements. Though FIG. **5** shows three of the PV modules **510**, **520**, **530** having different arrangements of the portions **512**, **522**, **532**, it will be apparent to those of skill in the art that any number of different arrangements are possible

without departing from the general concept embodied by the PV modules **510**, **520**, **530**.

Throughout this disclosure, the PV module **510** is indicated when it is desired to reference a PV module having color-contrasting material applied to the boundaries between some of the PV elements **500** so as to provide a non-uniform appearance as described above, but such reference to the PV module **510** is intended to refer to any of the PV modules **510**, **520**, **530**, or any other PV module having such features.

(47) In some embodiments, an exemplary solar roofing system includes wireways that are positioned between PV modules and are configured to enclose electrical cables that connect to the PV modules. In some embodiments, to facilitate providing an aesthetic appearance that is consistent both for the PV modules and the roofing shingles that form a solar roofing system, an exemplary solar roofing system includes wireways having a width that is matched to a width of the PV elements within the PV modules. FIGS. **6A**, **6B**, and **6C** show embodiments of an exemplary wireway **600** having an upper end **602**, a lower end **604**, a first side **606**, and a second side **608**. In some embodiments, the exemplary wireway **600** is configured to be installed on a roof deck such that the lower end **604** is at a lower elevation than is the upper end **602**. In some embodiments, the exemplary wireway **600** has a width as measured in a horizontal direction from the first side **606** to the second side **608** that is equal to an integer multiple of the PV cell width **237** for PV elements **236** that are used in the same solar roofing system as the exemplary wireway **600**. In some embodiments, the width of the wireway **600** is two times the PV cell width **237**. In some embodiments, the wireway **600** is rounded (e.g., so as to be concave on the side that faces the roof deck) to provide space to accommodate electrical cables and to soften the appearance of the wireway **600**. In some embodiments, the top surface of the wireway **600** (e.g., the side that faces away from the roof deck) includes at least one dark-colored portion and at least one light-colored portion.

(48) In some embodiments, the at least one light-colored portion includes a horizontal light-colored portion **610** extending across the wireway **600** from the first side **606** to the second side **608** adjacent the lower end **604**. In some embodiments, a height of the horizontal light-colored portion **610** (e.g., as measured in a vertical direction from the lower end **604** toward the upper end **602**) is equal to a creepage distance. As used herein, the creepage distance is the shortest distance along the surface of the insulating material between two conductive live parts or between conductive live parts and accessible part. For example, in embodiments detailed herein, the creepage distance is the shortest distance along the surface of an insulative portion of the PV module **510** between two conductive or accessible portions of the PV module **510**. In some embodiments, the creepage distance results in the appearance of a light-colored region along the long edges of the PV module **510**. Consequently, in some embodiments, the sizing of the horizontal light-colored portion **610** as equal to the creepage distance provides continuity of visual appearance between the wireway **600** and PV modules **510** that are adjacent thereto. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 10 millimeters to 30 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 12.5 millimeters to 27.5 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 15 millimeters to 25 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 17.5 millimeters to 22.5 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is about 20 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is 20 millimeters.

(49) In some embodiments, the at least one light-colored portion includes a vertical light-colored portion extending along the wireway **600** from the upper end **602** toward the lower end **604**. In some embodiments, as shown in FIG. **6A**, the vertical light-colored portion **612** extends along the wireway **600** and along the first side **606**. In some embodiments, as shown in FIG. **6B**, the vertical light-colored portion **614** extends along the wireway **600** and intermediate the first and second

sides **606**, **608**. In some embodiments, as shown in FIG. 6C, the vertical light-colored portion **616** extends along the wireway **600** and along the second side **608**. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is about equal to (e.g., within plus or minus 25%) the width of a cell gap, as described above with reference to FIG. 5. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is equal to the width of a cell gap. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is 6 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is about 6 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 5 mm to 7 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 4 mm to 8 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 3 mm to 9 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 3% of a width of the wireway **600** to 10% of the width of the wireway **600**. In some embodiments, the vertical light-colored portion **612** or **616** either provides a dark-colored region having a width that is twice the PV cell width **237**. In some embodiments, the vertical light-colored portion **614** provides two dark-colored regions, each of which has a width that is equal to the PV cell width **237**. Accordingly, in some embodiments, vertical light-colored portions **612**, **614**, **616** cause the wireway to have an appearance that is consistent with those of the roofing shingle **300** and the PV module **510**, thereby improving the consistency, visual flow, and aesthetic appearance of a roofing system including the wireway **600**.

(50) In some embodiments, an exemplary wireway **630**, as shown in FIG. 6D positioned between adjacent PV modules, is substantially similar to the wireway **600** described above with reference to FIGS. 6A-6C, other than as described hereinafter. In some embodiments, the wireway **630** has a flat top surface **632** and angled sides **634**, **636**. In some embodiments, the flat top surface **632** and angled sides **634**, **636** of the wireway **630** define a channel **638** within the wireway **630** to provide space to accommodate electrical cables. In some embodiments, the top surface of the wireway **630** (e.g., the side that faces away from the roof deck) includes at least one dark-colored portion and at least one light-colored portion. In some embodiments, the wireway **630** includes a horizontal light-colored portion **640** that is substantially similar to the horizontal light-colored portion **610** of the wireway **600** described above. In some embodiments, the wireway **630** includes a vertical light-colored portion **642**. The vertical light-colored portion **642** shown in FIG. 6D extends across the top surface **632** of the wireway **630** intermediate the sides **634**, **636**, in a manner similar to the vertical light-colored portion **614** shown in FIG. 6B, but it will be apparent to those of skill in the art that, in other embodiments, the vertical light-colored portion **642** of the wireway **630** may extend along either of the sides **634** or **636**, in a manner similar to the vertical light-colored portions **612** and **616** shown in FIGS. 6A and 6C, respectively.

(51) In some embodiments, an exemplary wireway **660**, as shown in perspective in FIG. 6E and in a front view in FIG. 6F, is substantially similar to the wireway **600** described above with reference to FIGS. 6A-6C, other than as described hereinafter. In some embodiments, the wireway **660** has a flat top **662** and sides **664**, **666** that are substantially perpendicular to the flat top **662**. In some embodiments, the flat top **662** and angled sides **664**, **666** of the wireway **660** define a channel **668** within the wireway **660** to provide space to accommodate electrical cables. In some embodiments, the top surface of the wireway **660** (e.g., the side that faces away from the roof deck) includes at least one dark-colored portion and at least one light-colored portion. In some embodiments, the wireway **660** includes a horizontal light-colored portion **670** that is substantially similar to the horizontal light-colored portion **610** of the wireway **600** described above. In some embodiments, the wireway **660** includes a vertical light-colored portion **672**. The vertical light-colored portion **672** shown in FIG. 6E extends across the top surface **632** of the wireway **630** along the side, in a manner similar to the vertical light-colored portion **612** shown in FIG. 6A, but it will be apparent to those of skill in the art that, in other embodiments, the vertical light-colored portion **672** of the wireway **660** may extend along the side **636**, in a manner similar to the vertical light-colored

portion **616** shown in FIG. **6C**, or may extend across the top surface **632** intermediate the sides **634**, **636**, in a manner similar to the vertical light-colored portion **614** shown in FIG. **6B**.

(52) In some embodiments, an exemplary roofing system includes a masking element applied to the border between adjacent PV modules. FIG. **7** shows an exemplary masking element **700** applied to the border between adjacent PV modules **710**, **720** (e.g., the PV modules **510**, **520**, **530** as described above with reference to FIG. **5**). In some embodiments, the masking element **700** is a dark color. In some embodiments, the masking element **700** is black. In some embodiments, the masking element **700** is made from a polymeric material. In some embodiments, the masking element **700** is made from a polymeric foam. In some embodiments, the masking element **700** is made from expanded polyethylene (“EPE”). In some embodiments, the masking element **700** is used in connection with the PV modules **710**, **720** that have a high-contrast backsheet **730**. In some embodiments, due to the high contrast of the backsheet **730**, any color difference between the masking element **700** and the PV modules **710**, **720** is swamped out, thereby hiding the boundary between the masking element **700** and the PV modules **710**, **720**. In some embodiments, by hiding the boundary between adjacent PV modules, use of the masking element **700** reduces the uniform, grid-like appearance of a roofing system including the PV modules **710**, **720**.

(53) In some embodiments, a solar roofing system including one or more of the roofing shingle **300**, the PV module **510**, the wireway **600**, and/or the masking element **700** provides an appearance that includes similar degrees and types of irregular/randomized appearance across all elements of the roofing system. In some embodiments, such a randomized appearance is harmonized around the cell pitch, i.e., the sum of the width of each PV cell and the cell gap. In some embodiments, such a similarly-randomized appearance causes the various elements of the solar roofing system to visually blend with one another, thereby providing a more aesthetically pleasing appearance to the overall solar roofing system

(54) FIG. **8A** shows an exemplary roofing system **800** that includes a plurality of the roofing shingle **300**, a plurality of the PV module **510**, a plurality of the wireway **600**, and a plurality of the masking element **700**. For clarity, only one of each of the elements noted above is specifically identified by a reference numeral in FIG. **8A**. FIG. **8B** shows a magnified view of a portion of the roofing system **800** including a plurality of the roofing shingle **300**, a plurality of the PV module **510**, and a plurality of the wireway **600** separating adjacent ones of the PV module **510**. FIG. **8C** shows a magnified view of a portion of the roofing system **800** including a plurality of the roofing shingle **300**, a plurality of the PV module **510**, and a plurality of the masking element **700** ones of the PV module **510** from adjacent ones of the roofing shingle **300**. FIG. **8D** shows a magnified view of a portion of the roofing system **800** including a plurality of the roofing shingle **300**, a plurality of the PV module **510**, and a plurality of the wireway **600** separating ones of the PV module **510** from adjacent ones of the roofing shingle **300**. FIG. **8E** shows a magnified view of a portion of the roofing system **800** including a plurality of the roofing shingle **300**, a plurality of the PV module **510**, and a plurality of the masking element **700** separating adjacent ones of the PV module **510**. It may be seen from FIGS. **8A-8E** that the roofing system **800** including a plurality of the roofing shingle **300**, a plurality of the PV module **510**, a plurality of the wireway **600**, and a plurality of the masking element **700** provides an aesthetic appearance in which the various elements of the roofing system **800** blend with one another aesthetically, rather than one in which the PV module **510** stands out. It will be apparent to those of skill in the art that the roofing system **800** shown in FIGS. **8A-8E** is only one exemplary manner of arranging a plurality of the roofing shingle **300**, a plurality of the PV module **510**, a plurality of the wireway **600**, and a plurality of the masking element **700**, and that any number of other arrangements of these same elements may be made. It will be further apparent to those of skill in the art that while the roofing system **800** shown in FIGS. **8A-8E** includes all of the roofing shingle **300**, the PV module **510**, the wireway **600**, and the masking element **700**, the same or similar aesthetic effect may be accomplished with a subset of these elements (including, but not limited to, with the roofing shingle **300** and the PV module **510**;

with the roofing shingle **300**, the PV module **510**, and the wireway **600**; or with the roofing shingle **300**, the PV module **510**, and the masking element **700**).

(55) In some embodiments, the various elements of the exemplary roofing system **800** mimic the water-shedding ability of a conventional roof shingle. In some embodiments, the various elements of the exemplary roofing system can be affixed to a roof deck using typical roofing methods such as nails or screws.

(56) In some embodiments, the roofing system **800** also includes at least one starter bar, a foot module, and a plurality of water shedding layers. In some embodiments, the roofing shingle **300** and/or the PV module includes an upper portion and a lower portion and is configured to be installed such that the upper portion is at a higher elevation than the lower portion. In some embodiments, the at least one starter bar is configured to be installed to a roof deck and includes a foot base. In some embodiments, a first one of the water shedding layers is configured to be installed over the foot base of the at least one starter bar, and at least one other one of the water shedding layers is configured to overlap and be installed over the first one of the plurality of water shedding layers. In some embodiments, the foot module is configured to be attached to the upper portion of the PV module **510** and/or the roofing shingle **300**. In some embodiments, the lower portion of the PV module **510** and/or the roofing shingle **300** is adapted to align with the foot base of the at least one starter bar, and the foot module is configured to be affixed to a last overlapping layer of the at least one of another of the first plurality of water shedding layers to the roof deck.

(57) While a number of embodiments of the present invention have been described, it is understood that these embodiments are illustrative only, and not restrictive, and that many modifications may become apparent to those of ordinary skill in the art. Further still, the various steps may be carried out in any desired order (and any desired steps may be added and/or any desired steps may be eliminated).

Claims

1. A roofing system, comprising: a plurality of photovoltaic modules on a roof deck, wherein each of the plurality of photovoltaic modules includes a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has one constant photovoltaic cell width; and a plurality of roofing shingles proximate to the plurality of photovoltaic modules on the roof deck, wherein each of at least some of the plurality of roofing shingles having a top surface, an exposure zone at a lower end of the top surface and a headlap zone at an upper end of the top surface, and a plurality of tooth portions extending from the lower end toward the headlap zone, wherein a first one of the plurality of tooth portions has a first width, wherein the first width is the photovoltaic cell width, wherein a second one of the plurality of tooth portions has a second width, wherein the second width is the photovoltaic cell width multiplied by a first positive integer that is greater than 1, wherein a third one of the plurality of tooth portions has a third width, wherein the third width is the photovoltaic cell width multiplied by a second positive integer that is greater than 1 and different than the first positive integer, wherein each of the roofing shingles does not include a photovoltaic cell.
2. The roofing system of claim 1, wherein the plurality of roofing shingles comprises at least a first roofing shingle, wherein the first roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt.
3. The roofing system of claim 2, wherein a top surface of the first roofing shingle comprises embedded granules.
4. The roofing system of claim 1, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is not identical to an arrangement of tooth portions of the first roofing shingle.

5. The roofing system of claim 1, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is identical to an arrangement of tooth portions of the first roofing shingle.
6. The roofing system of claim 1, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.
7. The roofing system of claim 1, wherein the plurality of tooth portions of at least some of the plurality of roofing shingles comprises a fourth one of the plurality of tooth portions having a fourth width, wherein the fourth width is the photovoltaic cell width multiplied by 0.5.
8. A roofing system, comprising: a plurality of photovoltaic modules on a roof deck, wherein each of the plurality of photovoltaic modules includes a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has one constant photovoltaic cell width; and a plurality of roofing shingles proximate to the plurality of photovoltaic modules on the roof deck, wherein each of at least some of the plurality of roofing shingles having a top surface, an exposure zone at a lower end of the top surface and a headlap zone at an upper end of the top surface, and a plurality of tooth portions extending from the lower end toward the headlap zone, wherein a first one of the plurality of tooth portions has a first width, wherein the first width is the photovoltaic cell width multiplied by a first positive integer that is greater than 1, wherein a second one of the plurality of tooth portions has a second width, wherein the second width is the photovoltaic cell width multiplied by a second positive integer that is greater than 1 and different than the first positive integer, wherein each of the roofing shingles does not include a photovoltaic cell.
9. The roofing system of claim 8, wherein the plurality of roofing shingles comprises at least a first roofing shingle, wherein the first roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt.
10. The roofing system of claim 9, wherein a top surface of the first roofing shingle comprises embedded granules.
11. The roofing system of claim 8, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is not identical to an arrangement of tooth portions of the first roofing shingle.
12. The roofing system of claim 8, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is identical to an arrangement of tooth portions of the first roofing shingle.
13. The roofing system of claim 8, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.
14. The roofing system of claim 8, wherein the plurality of tooth portions of at least some of the plurality of roofing shingles comprises a third one of the plurality of tooth portions having a third width, wherein the third width is the photovoltaic cell width multiplied by 0.5.
15. A method, comprising: obtaining a plurality of photovoltaic modules, wherein each of the plurality of photovoltaic modules includes a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has one constant photovoltaic cell width; and obtaining a plurality of roofing shingles, wherein each of at least some of the plurality of roofing shingles having a top surface, an exposure zone at a lower end of the top surface and a headlap zone at an upper end of the top surface, and a plurality of tooth portions from the lower end toward the headlap zone, wherein a first one of the plurality of tooth portions has a first width, wherein the first width is the photovoltaic cell width, wherein a second one of the plurality of tooth portions has a second width, wherein the second width is the photovoltaic cell width multiplied by a first positive integer that is greater than 1, wherein a third one of the plurality of tooth portions has a third width, wherein the third width is the photovoltaic cell width multiplied by a second positive integer that is greater than 1 and different than the first positive integer, wherein each of the roofing shingles do not include a photovoltaic cell; and installing the plurality of photovoltaic modules and the plurality

of roofing shingles proximate one another above a roof deck.

16. The method of claim 15, wherein the plurality of roofing shingles comprises at least a first roofing shingle, wherein the first roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt.

17. The method of claim 15, wherein a top surface of the first roofing shingle comprises embedded granules.

18. The method of claim 15, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.

19. The method of claim 15, wherein the plurality of tooth portions of at least some of the plurality of roofing shingles comprises a fourth one of the plurality of tooth portions having a fourth width, wherein the fourth width is the photovoltaic cell width multiplied by 0.5.

20. The method of claim 19, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.
